Proficient in SketchUp (80 Hours)

S.No	Session	Topics Covered	Type	Duration (hrs)
1	Session 1	**Introduction** Getting Started with SketchUp Understanding the Template **GUI** Understanding SketchUp Interface Understanding Various Menu Understanding the Status Bar Understanding the Default Tray Understanding the Toolbars Customization Managing Trays Creating Shortcut key **Mouse Control & SketchUp preferences** Creating Saving Opening a model Importing and Exporting Different Types file of SketchUp **Axes** Understanding the Axes in SketchUp Reorienting the Axes **Inference** Concept of Inference Understanding the Various inference types Locking the Inference	Theory/Demo	2.0 Hrs
2	Session 2	Practice 01: Importing Different Types of file into SketchUp	Practical	2.0 Hrs
3	Session 3	**Modelling Of Entities 1** **Drawing Tool** Understanding the line tool Constructing a surface Divide and Joining a face Understanding the various input method **Rectangle** Understanding the rectangle tool and various dimensions inputs Drawing a Rotated rectangle entity. Understanding the creation of circle entity and its various parameters **Ellipse** Drawing an ellipse from circle Understanding the creation of Polygon entity and its various parameters **Arc** Drawing an Arc using various Arc tools 2 point Arc 3 point Arc Pie tool Drawing a freehand Shapes **Selection of an object** Understanding the Select tool Selecting multiple entities Adding and subtracting from a selection set.	Theory/Demo	2.0 Hrs
4	Session 4	Practice02: Create a simple 2D floor plan using the drawing tools	Practical	2.0 Hrs
5	Session 5	**Modelling Of Entities 2** **Modify Tools** Editing Shapes using Entity Info option Pushing and pulling shapes into 3D Cutting a 3D Shape out of a model Understanding the move tool Stretching geometry using Move tool Resizing Surface Entities Creating Copies using Move Tool and Rotate Tool Creating Multiple Copies/Array using Move tool and Rotate Tool Offsetting Line from an existing geometry. **Undo recent & Eraser tool**	Theory/Demo	2.0 Hrs

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6	Session 6	Practice 03: Create a 3D floor plan using modify tools.	Practical	2.0 Hrs
7	Session 7	**Modelling Of Entities 3** **Modify Tools** Understand the Dividing, Splitting and Exploding Entities Understand the Flipping, Mirroring and Rotating Scaling a model using Scale tool Stretching a model using scale tool Understand the Follow Me Tool Automatic and Manual Extrusion with Follow Me Tool Modelling Revolved Shape, Soften/ Smooth Edges of Geometry & Hiding Geometry **Tape Measure Tool & Protractor Tool** Measuring Distance/ Create guides Measuring an angle Editing Guidelines	Theory/Demo	2.0 Hrs
8	Session 8	Practice 04: Create Vase and Sun Shade using Follow Me	Practical	2.0 Hrs
9	Session 9	**Modelling Of Entities 4** **Annotative Model** Understanding the Annotation tools Adding Text Text as a Model Entity Dimension Tool **Matching a photo to a model** Match Photo's Colour- coded tools Creating a 3D Model from a photo Deleting a matched photo	Theory/Demo	2.0 Hrs
10	Session 10	Practice 05: Apply Text, Dimension to the model and Create building model using Matching photo.	Practical	2.0 Hrs
11	Session 11	**Groups** **Grouping Geometry** Working with hierarchies in the outliners Renaming groups and components Finding groups and components Identifying a group/Components Restructuring a model's hierarchy	Theory/Demo	2.0 Hrs
12	Session 12	Practice 06: Create Table in Group using Follow Me and Revolving tool.	Practical	2.0 Hrs
13	Session 13	**Components** **Getting Started with Components** Understanding the components Creating a component Editing a component Editing all instances of a component **Making Dynamic component** Adding attributes Defining Attribute Values and testing Inserting, replacing and Interacting with dynamic Component Searching for component Adjusting a component's inserting point Editing component. Importing and Exporting SketchUp Model into 3D warehouse	Theory/Demo	2.0 Hrs
14	Session 14	Practice 07: Create Door in Components and Modify its properties.	Practical	2.0 Hrs
15	Session 15	**Sandbox and Solid Tools** **Sandbox** Create a terrain from edge selection Create a terrain Mesh Flat foundation Project Add Edges Smoothness Loft or depress terrain faces **Solid Tools** Outer Shell	Theory/Demo	2.0 Hrs

		Union		
		Intersect		
		Subtract		
		Trim		
		Split		
16	Session 16	Practice 08: Create contour Landscape using Sandbox tool	Practical	2.0 Hrs
17	Session 17	**1001 Bit Pro - Plugins**	Theory/Demo	2.0 Hrs
		Staircase		
		Window		
		Wall		
		Door		
		Openings on walls		
		Extend		
18	Session 18	Practice 09: Create cantilever Dog leg staircase Components using 1001 plugin	Practical	2.0 Hrs
19	Session 19	**1001 Bit Pro - Plugins**	Theory/Demo	2.0 Hrs
		Roof		
		Rafter		
		Purlins		
		Chamfer		
		Fillet		
20	Session 20	Practice 10: Create Slope roof using 1001 plugin	Practical	2.0 Hrs
21	Session 21	**Architect Tool and True Bend - Plugins**	Theory/Demo	2.0 Hrs
		Architect Tool		
		Generate Buildings		
		Merge Solid Buildings		
		Fill Solid Holes		
		Select Non-Solids		
		Make 2:1 Road Profile		
		Move to Z		
		Contour Tool		
		Extrude Up		
		Project Down Tool		
		Magnet Tool		
		Move to Plane		
		Flatten Selection		
		Crop Selection to Boundary		
		Grid Divide		
22	Session 22	Practice 11: create a neighbourhood buildings using architect tool.	Practical	2.0 Hrs
23	Session 23	**Architecture Design - Exterior**	Theory/Demo	2.0 Hrs
		SketchUp FFD		
		2X2 FFD, 3X3 FFD and NXN FFD		
		Lock and unlock Edge		
		Make patch		
		Flowify		
		Exploring Architecture design and form development with demo		
		True Bend		
24	Session 24	Practice 12: Recreate a Architecture reference building in SketchUp	Practical	2.0 Hrs
25	Session 25	**Presenting Your Model**	Theory/Demo	2.0 Hrs
		Viewing a Model		
		Camera		
		Changing the view or focal length		
		Section		
		To place a section plane		
		Filling voids in section cuts		
		Creating new geometry from a section		
		Showing/hiding section planes		
		Exporting section cut effects		
		Style		
		Choosing a style		
		Editing Styles		
		Style Editing options		
		Managing in-model styles and collections		

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		Sharing styles		
		Adding fog effects to model		
26	Session 26	Practice 13: Apply style and Create Scenes of SketchUp Model using Camera.	Practical	2.0 Hrs
27	Session 27	**Presenting Your Model** **Applying Material & Texture** Adding colour and Textures with material Applying Material & Texture Replacing one material with another Editing Material-Editing Option Calculating Material area Managing and organizing material Creating new material Applying Texture Adding Geographic location from Google Map Applying Real-time shadows.	Theory/Demo	2.0 Hrs
28	Session 28	Practice 14: Apply Material and texture to the model. Create new material to the model.	Practical	2.0 Hrs
29	Session 29	**Advance Model in SketchUp** **Interior Design** Definition and process of interior design Design principle and Elements Introduction to the design of interior spaces Design Concept Colour schemes Properties of colour Types of colour Mood Board - creation	Theory/Demo	2.0 Hrs
30	Session 30	Practice 15: Create a mood board for the residence Living Room ,office ,reception area and present it	Practical	2.0 Hrs
31	Session 31	**COMPONENTS OF INTERIOR SPACE- LIGHTING** **Lighting** Introduction to interior lighting Different types of lighting Types of lighting fixtures Lights physical properties and effects on spaces Modelling a interior and exterior light component.	Theory/Demo	2.0 Hrs
32	Session 32	Practice 16: Create a Chandelier light component .	Practical	2.0 Hrs
33	Session 33	**COMPONENTS OF INTERIOR SPACE- FURNITURES** **Furniture's** Introduction to interior Furniture Furniture related to human comfort and function Types of Furniture Furniture properties and effects on spaces Modelling a Living room Furniture component.	Theory/Demo	2.0 Hrs
34	Session 34	Practice 17: Create a tea Table and sofa component.	Practical	2.0 Hrs
35	Session 35	**COMPONENTS OF INTERIOR SPACE- FURNITURES** Modelling a Bedroom room Furniture component. **Cloth work** Quadrilateral, Equilateral, and Adaptive grids Split Edges and weld Edges Loop Subdivision and Laplacian Smoothing Drape and Undrape Record and Export	Theory/Demo	2.0 Hrs
36	Session 36	Practice 18: Create a Bed , side table and curtain component .	Practical	2.0 Hrs
37	Session 37	**Landscape** Introduction to landscape architecture Hard and soft landscape elements Urban landscape Landscape and garden design Modelling a Residence garden landscape.	Theory/Demo	2.0 Hrs
38	Session 38	Practice 19: Create a Back yard Garden landscape using Sand box tool.	Practical	2.0 Hrs
		3D Model to 2D		

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39	Session 39	**SketchUp Layout** Introduction Template Selection Understanding the GUI Creating Documents in Layout **Presenting layout Documents** Scale Changing Views Changing Model Style Changing Dimension Style Inserting Text & Changing Text Style Exporting/ printing Layout Document	Theory/Demo	2.0 Hrs
40	Session 40	Practice 20: Convert your 3D SketchUp to 2D layout	Practical	2.0 Hrs